

Support Vector Machine

Classification and Regression

Equation of Line, 2D Plane and Hyperplane

Equation of Line

Slope-Intercept Form

$$y = mx + c$$

where

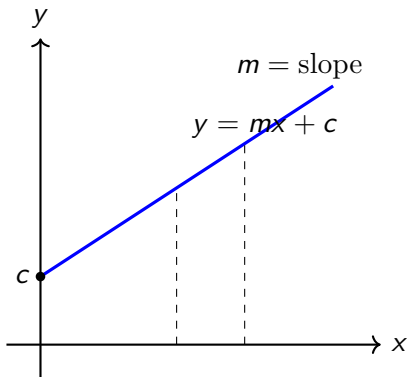
- m : Slope
- c : Y-intercept

It can also be written as

$$y = \beta_0 + \beta_1 x$$

where

$$\beta_1 = m, \quad \beta_0 = c$$



General Form of a Line

The general equation of a line is

$$ax + by + c = 0$$

Because every straight line in a 2D plane can be written in the above form.
Rearranging,

$$by = -ax - c$$

Dividing by b ,

$$y = -\frac{a}{b}x - \frac{c}{b}$$

Comparing with

$$y = mx + c,$$

we obtain

$$m = -\frac{a}{b}, \quad c = -\frac{c}{b}$$

Equation of a Plane (Decision Boundary)

For two input features x_1 and x_2 , the linear equation becomes

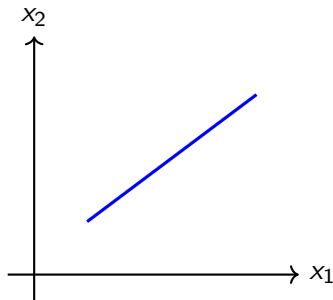
$w_1x_1 + w_2x_2 + b = 0$ where

- w_1 and w_2 = weight corresponding to feature x_1 and x_2
- b = bias (intercept)

This equation represents a straight line in a two-dimensional feature space.

$w_1x_1 + w_2x_2 + b = 0$ or equivalently, $x_2 = -\frac{w_1}{w_2}x_1 - \frac{b}{w_2}$ Hence,

Slope = $-\frac{w_1}{w_2}$, Intercept = $-\frac{b}{w_2}$.



Hyperplane Equation and Representation

General Hyperplane Equation

For an n -dimensional feature space,

$$w_1x_1 + w_2x_2 + \cdots + w_nx_n + b = 0$$

or equivalently,

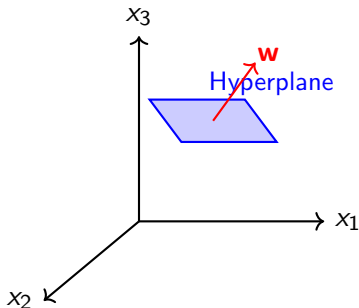
$$\mathbf{w}^T \mathbf{x} + b = 0$$

where

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The dot product is

$$\mathbf{w}^T \mathbf{x} = w_1x_1 + w_2x_2 + \cdots + w_nx_n$$



$$\mathbf{w}^T \mathbf{x} + b = 0$$

Interpretation

A hyperplane is a flat decision boundary that divides the feature space into two regions or classes. The vector $\mathbf{w}$ is perpendicular (normal) to the hyperplane, while b determines its position relative to the origin. The hyperplane acts as the **decision boundary** in machine learning algorithms such as Logistic Regression, Perceptron, and Support Vector Machines.

Hyperplane in Different Dimensions

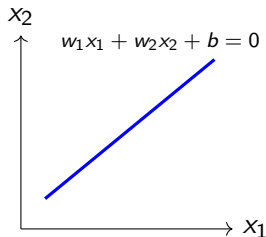
Dimension	Equation	Representation
1-D	$w_1x + b = 0$	Point
2-D	$w_1x_1 + w_2x_2 + b = 0$	Line
3-D	$w_1x_1 + w_2x_2 + w_3x_3 + b = 0$	Plane
n -D	$\mathbf{w}^T \mathbf{x} + b = 0$	Hyperplane

Key Points

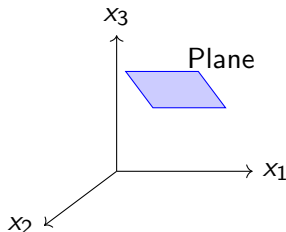
- $\mathbf{w}$ is called the **weight (coefficient) vector**.
- $\mathbf{x}$ is the **feature vector**.
- b is the **bias (intercept)**.
- A hyperplane is an $(n - 1)$ dimensional flat surface that divides an n -dimensional feature space into two regions.

Geometric Representation of a Hyperplane

For two features,
 $w_1x_1 + w_2x_2 + b = 0$
represents a Straight Line



For three features,
 $w_1x_1 + w_2x_2 + w_3x_3 + b = 0$
represents a Plane



In Support Vector Machine, our objective is not just to find any hyperplane, but to find the optimal hyperplane that best separates the two classes by maximizing the margin

Normal Vector of a Hyperplane

The hyperplane is

$$\mathbf{w}^T \mathbf{x} + b = 0$$

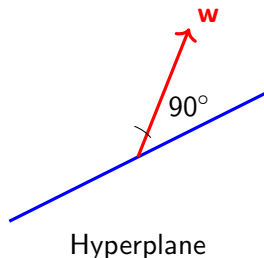
The vector

$$\mathbf{w}$$

is called the Normal Vector because it is perpendicular to the hyperplane.

For any point $\mathbf{x}$ on the hyperplane, $\mathbf{w}^T \mathbf{x} + b = 0$.

The shortest distance from any point to the hyperplane is measured along the direction of the normal vector $\mathbf{w}$.



Distance Between a Point and a Hyperplane

Hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$

Point

$$\mathbf{x}_0 = \begin{bmatrix} x_{01} \\ x_{02} \\ \vdots \\ x_{0n} \end{bmatrix}$$

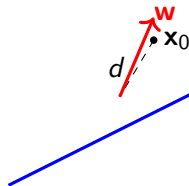
The perpendicular distance is

$$d = \frac{|\mathbf{w}^T \mathbf{x}_0 + b|}{\|\mathbf{w}\|}$$

where $\|\mathbf{w}\| = \sqrt{w_1^2 + w_2^2 + \cdots + w_n^2}$.

If $\mathbf{w}^T \mathbf{x}_0 + b > 0$, the point lies on the positive side.

If $\mathbf{w}^T \mathbf{x}_0 + b < 0$, the point lies on the negative side.



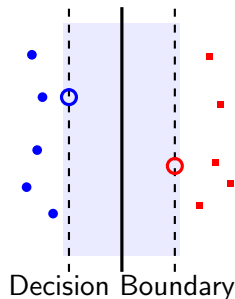
In SVM, the perpendicular distance is computed for every training point from the decision boundary. The nearest points are called **support vectors**, and they determine the position of the optimal hyperplane.

Support Vector Machine (SVM)

Definition

A **Support Vector Machine (SVM)** is a supervised machine learning algorithm used for both classification and regression. Its primary objective is to find the **optimal separating hyperplane** that maximizes the distance (margin) between different classes.

- Mainly used for binary classification
- Uses **support vectors** (critical data points) to find the **maximum margin hyperplane**.
- Among all possible separating hyperplanes, SVM selects the one that leaves the **largest possible margin** between the two classes, leading to better generalization on unseen data.

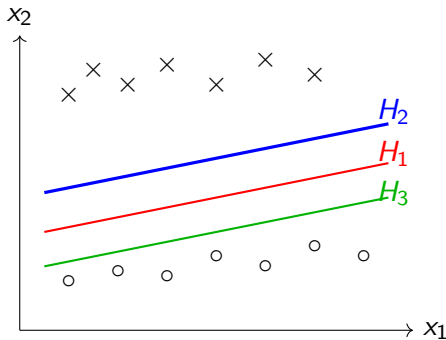


Why Do We Need Support Vector Machines?

Suppose we have a binary classification problem.

- A linearly separable dataset may have infinitely many separating hyperplanes.
- Although all hyperplanes correctly classify the training data, they do not provide the same generalization performance.

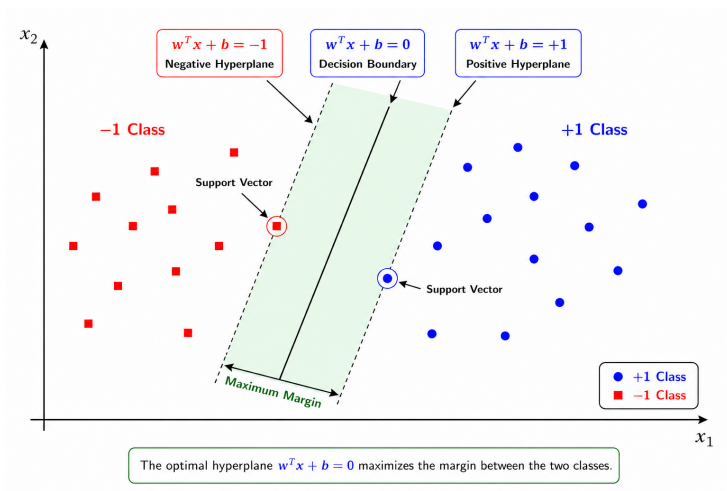
Which hyperplane should we choose?



Which Hyperplane is the Best?

- All three hyperplanes correctly classify the training data.
- However, their ability to classify unseen data differs.
- **Support Vector Machine (SVM)** selects the hyperplane with the **maximum margin**.
- This principle is called the **Maximum Margin Classifier**.

Maximum Margin Hyperplane



The maximum margin hyperplane is equidistant from the nearest training samples of both classes.

Linearly Separable Data

- Blue circles belong to Class +1
- Red squares belong to Class -1
- Many separating lines are possible.
- SVM chooses the one with the **maximum margin**.

Equation of the Margin Hyperplanes

The two margin hyperplanes are

$$\mathbf{w}^T \mathbf{x} + b = 1$$

and

$$\mathbf{w}^T \mathbf{x} + b = -1$$

The decision boundary lies exactly midway between them.

$$\mathbf{w}^T \mathbf{x} + b = 0$$

Hence,

Margin Width = Distance between the two margin hyperplanes.

Hyperplane Equation

For an n -dimensional feature space,

$$w^T x + b = 0$$

where

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$f(x) = w^T x + b$$

Decision rule

$$f(x) > 0 \Rightarrow +1$$

$$f(x) < 0 \Rightarrow -1$$

Functional Margin

Suppose the class labels are

$$y_i \in \{-1, +1\}$$

For correct classification,

$$y_i(w^T x_i + b) > 0$$

SVM scales the parameters since there are infinitely many values of w and b that represent the same decision boundary. so

$$y_i(w^T x_i + b) \geq 1$$

Hence, Positive support hyperplane

$$w^T x + b = 1$$

Negative support hyperplane

$$w^T x + b = -1$$

Decision boundary

$$w^T x + b = 0$$

Derivation of Margin

Distance from a point to a hyperplane

$$d = \frac{|w^T x + b|}{\|w\|}$$

Distance from the support hyperplane $w^T x + b = 1$ to the decision boundary $w^T x + b = 0$ is

$$\frac{1}{\|w\|}$$

Similarly, distance from $w^T x + b = -1$ to $w^T x + b = 0 = \frac{1}{\|w\|}$

Therefore,

$$\text{Margin} = \frac{2}{\|w\|}$$

Optimization Problem

Maximum margin

$$\max \frac{2}{||w||}$$

is equivalent to

$$\min ||w||$$

For mathematical convenience,

$$\min \frac{1}{2} ||w||^2$$

subject to

$$y_i(w^T x_i + b) \geq 1$$

This is called the **Primal Optimization Problem**.

Need for Lagrange Multipliers

SVM is a **constrained optimization problem**. The objective is to minimize

$$\frac{1}{2} \|w\|^2$$

while satisfying the constraint

$$y_i(w^T x_i + b) \geq 1, \quad i = 1, 2, \dots, n.$$

Since ordinary differentiation cannot handle constraints directly, the **Lagrange Multiplier Method** combines the objective function and the constraints into a single **Lagrangian function**, which can then be optimized using differentiation.

Using Lagrange Multipliers

Objective:

$$\min \frac{1}{2} \|w\|^2$$

Constraint:

$$y_i(w^T x_i + b) \geq 1, \quad i = 1, 2, \dots, n$$

Introduce a Lagrange multiplier $\alpha_i \geq 0$ for each constraint and combine the objective and constraints into the Lagrangian:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i(w^T x_i + b) - 1]$$

$$\min_{w, b} \max_{\alpha \geq 0} L(w, b, \alpha)$$

Result: Minimize with respect to w, b and maximize with respect to α_i to obtain the optimal maximum-margin hyperplane.

Derivative with Respect to w

Take partial derivative

$$\frac{\partial L}{\partial w} = 0$$

Therefore,

$$w - \sum_i \alpha_i y_i x_i = 0$$

Hence,

$$w = \sum_i \alpha_i y_i x_i$$

The weight vector is a linear combination of training samples.

Derivative with Respect to b

Take derivative

$$\frac{\partial L}{\partial b} = 0$$

$$-\sum_i \alpha_i y_i = 0$$

Hence,

$$\boxed{\sum_i \alpha_i y_i = 0}$$

This is another optimization constraint.

Dual Optimization Problem (Step 1)

Substitute

$$w = \sum_i \alpha_i y_i x_i$$

into the Lagrangian

$$L = \frac{1}{2} w^T w - \sum_i \alpha_i y_i w^T x_i + \sum_i \alpha_i$$

Since

$$\sum_i \alpha_i y_i = 0,$$

the term containing b becomes zero.

Dual Optimization Problem (Step 2)

Compute

$$\begin{aligned}w^T w &= \left(\sum_i \alpha_i y_i x_i \right)^T \left(\sum_j \alpha_j y_j x_j \right) \\&= \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j\end{aligned}$$

Similarly,

$$\sum_i \alpha_i y_i w^T x_i = \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j.$$

Substitute these expressions into the Lagrangian.

Dual Optimization Problem (Step 3)

The Lagrangian becomes

$$L = \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j - \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum_i \alpha_i$$

Simplifying,

$$L = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j$$

Hence, the dual optimization problem is

$$\max_{\alpha} \left(\sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j \right)$$

subject to

$$\alpha_i \geq 0, \quad \sum_i \alpha_i y_i = 0.$$

Constraints of the Dual Optimization Problem

The dual optimization problem is subject to constraints $\alpha_i \geq 0$

$$\sum_{i=1}^m \alpha_i y_i = 0$$

Interpretation of the Constraints

- $\alpha_i \geq 0$:
 - Every Lagrange multiplier must be zero or positive.
 - If $\alpha_i = 0$, the corresponding training sample does not influence the classifier.
 - If $\alpha_i > 0$, the corresponding training sample becomes a **support vector**.

- $\sum_{i=1}^m \alpha_i y_i = 0$:
 - Balances the contributions of the positive and negative classes.
 - Ensures that the resulting hyperplane correctly separates the two classes.

Dual Optimization Problem

After solving the dual optimization problem,

- The **primal optimization problem** is transformed into the **dual optimization problem**.
- The optimization is performed only over the **Lagrange multipliers** α_j .
- Only training samples with

$$\alpha_j > 0$$

become the **support vectors**.

- The **support vectors** determine the position of the optimal separating hyperplane.
- The dual formulation depends only on the inner products

$$x_i^T x_j,$$

which enables the use of **kernel functions** for non-linear classification.

Decision Function

We know,

$$w = \sum_i \alpha_i y_i x_i$$

Only points with

$$\alpha_i > 0$$

are called **Support Vectors**. Classifier

$$f(x) = \sum_i \alpha_i y_i x_i^T x + b$$

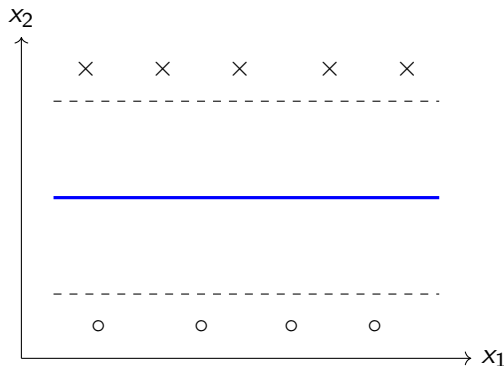
Prediction

$$\hat{y} = \text{signum}(f(x))$$

$$\text{signum}(f(x)) = \begin{cases} +1, & \text{if } f(x) > 0, \\ 0, & \text{if } f(x) = 0, \\ -1, & \text{if } f(x) < 0. \end{cases}$$

Hard Margin Support Vector Machine

Hard Margin SVM assumes that the dataset is perfectly linearly separable.

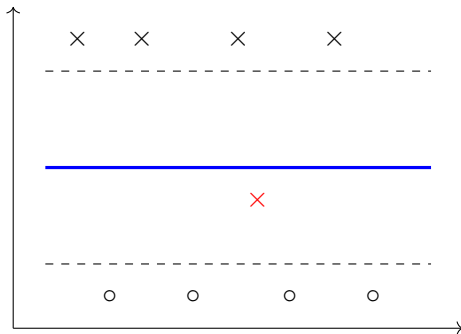


Characteristics

- No misclassification allowed.
- Works only for perfectly separable datasets.
- Very sensitive to noise and outliers.

Limitation of Hard Margin SVM

Real-world datasets usually contain noisy observations.



A single noisy point may make the data non-separable.
Hence, Hard Margin SVM fails.

Soft Margin Support Vector Machine

Soft Margin SVM allows a few samples to violate the margin.

Instead of enforcing

$$y_i(w^T x_i + b) \geq 1,$$

it introduces a slack variable

$$\xi_i \geq 0$$

which measures the degree of margin violation.

Soft Margin provides better generalization for noisy datasets.

Slack Variables in Soft-Margin SVM

Soft-Margin SVM Constraint

$$y_i (w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

The slack variable measures how much a sample violates the margin.

Slack Variable (ξ_i)

- $\xi_i = 0$ Correctly classified, on or outside the margin
- $0 < \xi_i < 1$ Correctly classified, inside the margin.
- $\xi_i = 1$ On the decision boundary.
- A larger value of ξ_i indicates a greater violation of the margin.
- Slack variables allow SVM to tolerate noisy or overlapping data while still maximizing the margin.

Soft Margin Optimization

The optimization problem becomes

$$\min \left(\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \right)$$

Subject to

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

where

$$\xi_i \geq 0.$$

The first term maximizes the margin.

The second term penalizes classification errors.

Meaning of Parameter C

- The parameter C controls the trade-off between Large margin and Training accuracy
- Large Margin: Better Generalization and allow few classification.
- Small Margin: High training accuracy and more prone to overfitting.

Large $C \rightarrow$ Less misclassifications

Small $C \rightarrow$ Larger margin

Hinge Loss Function

Unlike Logistic Regression, SVM uses the Hinge Loss.

$$L = \max(0, 1 - yf(x))$$

where

$$f(x) = w^T x + b.$$

Properties

- Zero loss for correctly classified samples outside the margin.
- Positive loss for samples inside the margin.
- Linear penalty for misclassified samples.

Hard Margin vs Soft Margin

Hard Margin	Soft Margin
Perfectly separable data	Noisy data
No misclassification	Allows misclassification
Sensitive to outliers	Robust to outliers
No slack variables	Uses slack variables
Limited practical use	Widely used in practice

Summary

- Hard Margin SVM assumes perfectly separable data.
- Soft Margin SVM introduces slack variables.
- Parameter C controls the trade-off between margin and classification error.
- Hinge Loss is the objective function used by SVM.
- Soft Margin SVM is preferred in most practical applications.